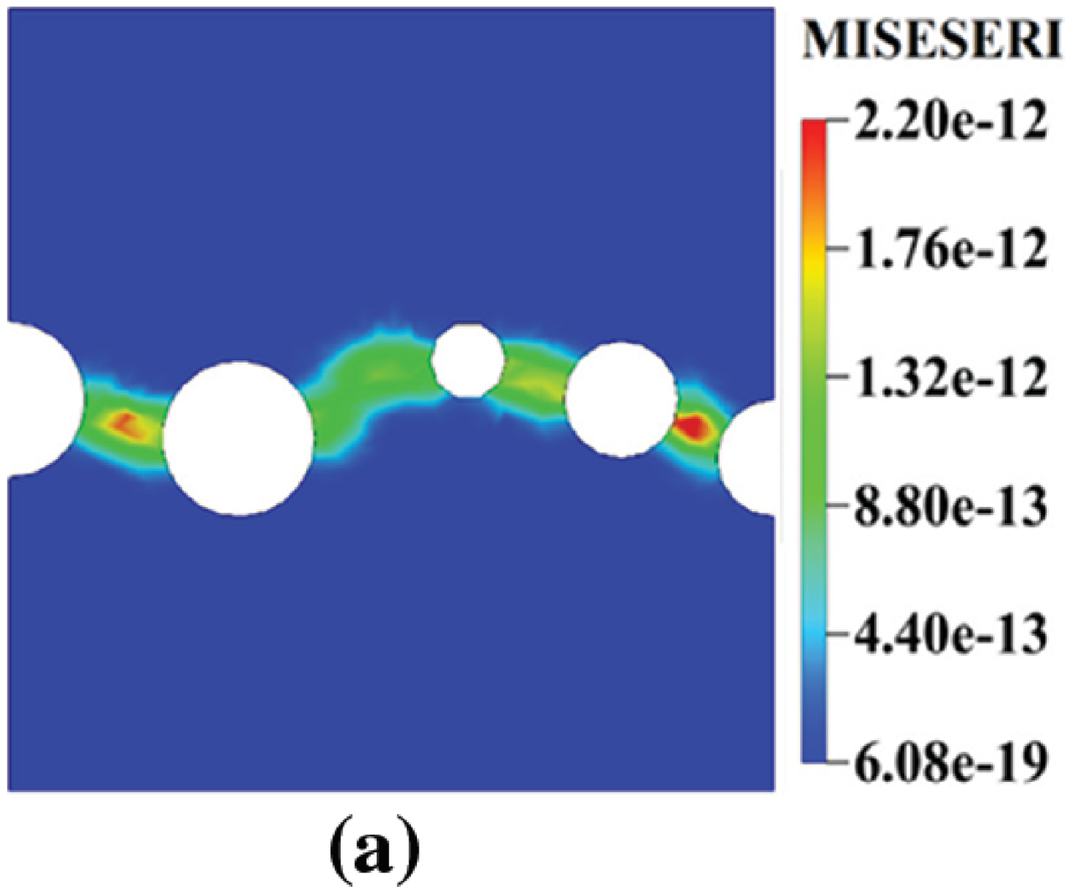
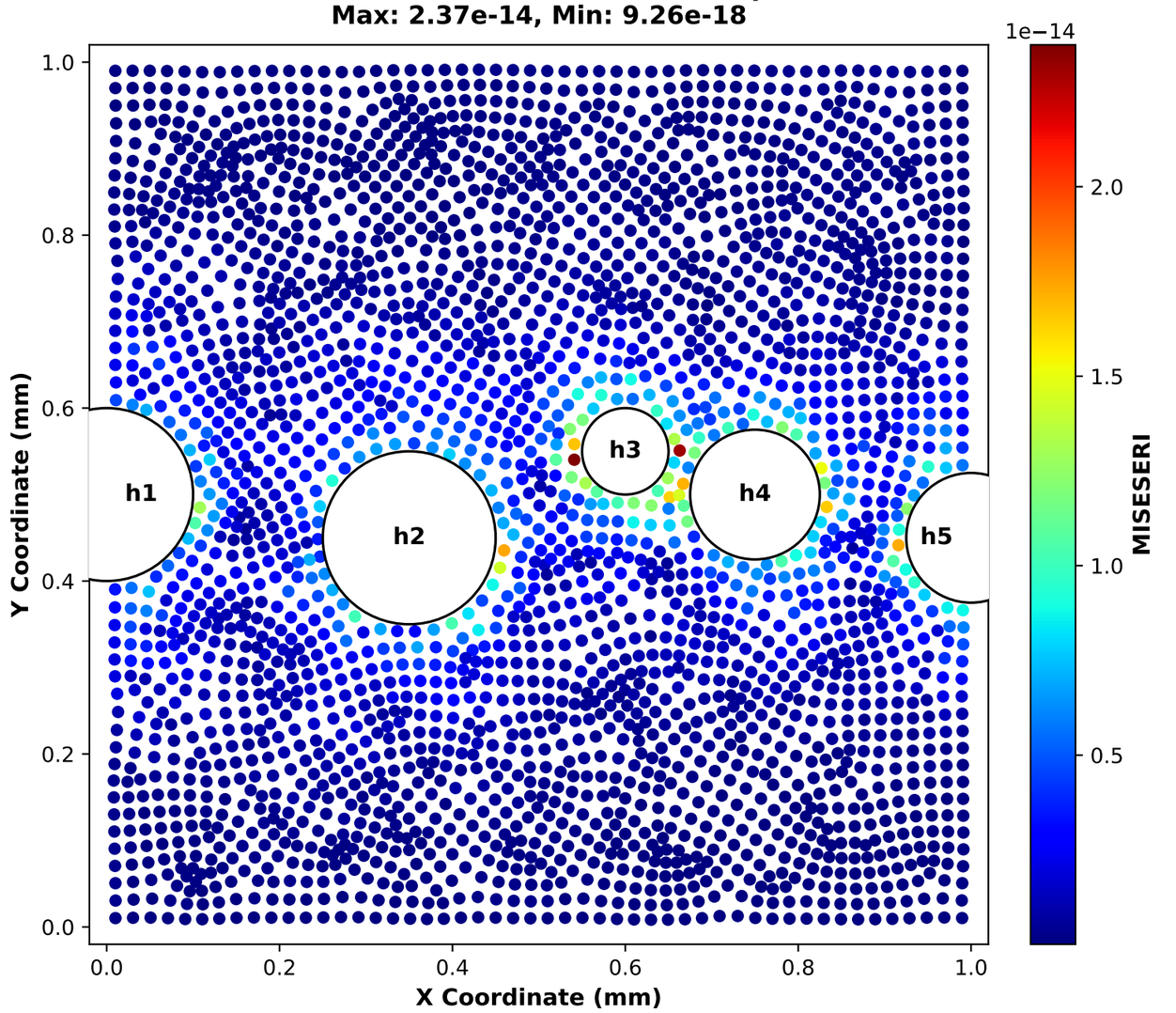
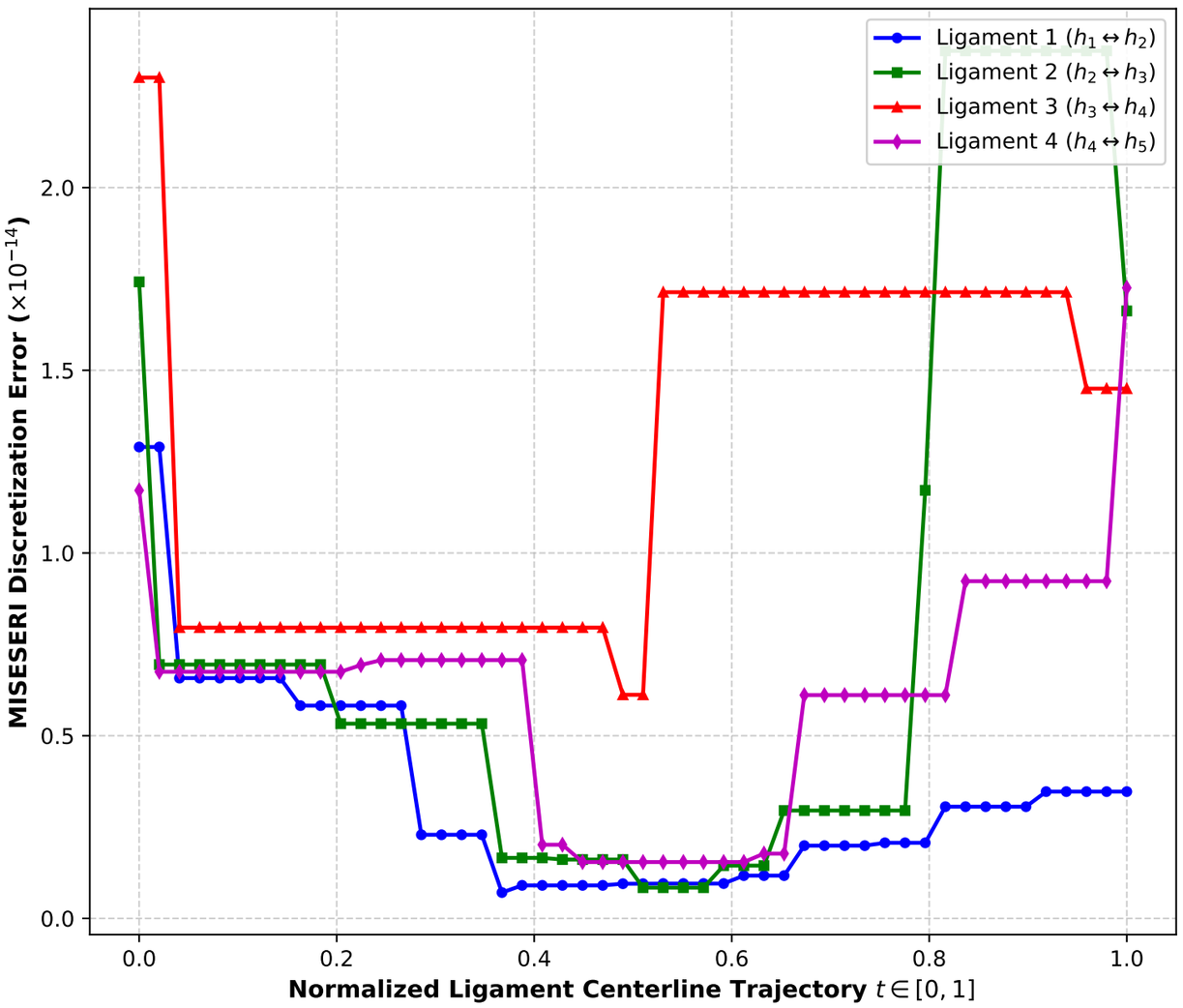
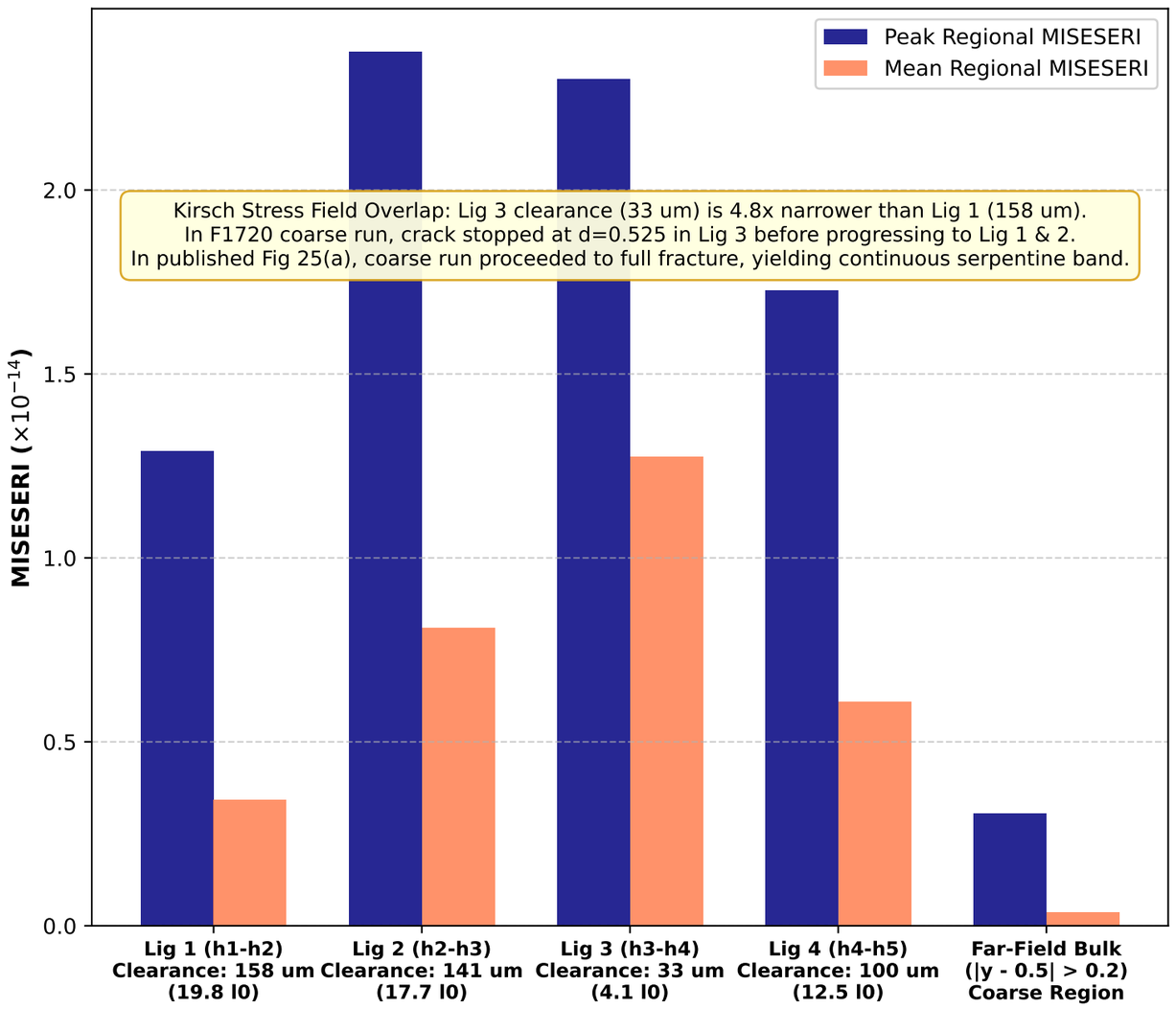

(a) Published Pandey & Kumar (2025) Figure 25(a)
Native MISESERI [Max: 2.20e-12, Min: 6.08e-19]
MISESERI
2.20e-12
1.76e-12
1.32e-12
8.80e-13
4.40e-13
6.08e-19
(a)


(b) Authentic F1720 Coarse Donor Field (Step-2, t=0.7257)
Max: 2.37e-14, Min: 9.26e-18
1e−14
MISESERI
Y Coordinate (mm)
X Coordinate (mm)
h1
h2
h3
h4
h5


(c) Inter-Hole Centerline Error Comparison
Ligament 1 ($h_1 \leftrightarrow h_2$)
Ligament 2 ($h_2 \leftrightarrow h_3$)
Ligament 3 ($h_3 \leftrightarrow h_4$)
Ligament 4 ($h_4 \leftrightarrow h_5$)
MISESERI Discretization Error ($\times 10^{-14}$)
Normalized Ligament Centerline Trajectory $t \in [0, 1]$


(d) Regional Error Disparity & Geometric Clearance Decay
Peak Regional MISESERI
Mean Regional MISESERI
Kirsch Stress Field Overlap: Lig 3 clearance (33 um) is 4.8x narrower than Lig 1 (158 um).
In F1720 coarse run, crack stopped at d=0.525 in Lig 3 before progressing to Lig 1 & 2.
In published Fig 25(a), coarse run proceeded to full fracture, yielding continuous serpentine band.
MISESERI ($\times 10^{-14}$)
Lig 1 (h1-h2) Clearance: 158 um (19.8 l0)
Lig 2 (h2-h3) Clearance: 141 um (17.7 l0)
Lig 3 (h3-h4) Clearance: 33 um (4.1 l0)
Lig 4 (h4-h5) Clearance: 100 um (12.5 l0)
Far-Field Bulk (|y - 0.5| > 0.2) Coarse Region